

5.7.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers in the microbiome sequencing services market is moderate. Academic research institutes (one of the largest end users of sequencing products) are involved in bulk purchases of products and bulk sequencing of samples. This trend of bulk purchasing further increases the bargaining power of buyers in this market.

5.7.5 INTENSITY OF COMPETITIVE RIVALRY

The microbiome sequencing services market is fragmented in nature, with the top 3 players accounting for a share of ~20%. A large number of local players and small-scale CROs account for the remaining market share. These players operate in niche market segments. Industry players are expected to cash in on the various growth opportunities and gain maximum share in the market. This has created low-entry barriers for new players, and the existing players compete to expand their shares.

5.8 REGULATORY ANALYSIS

5.8.1 NORTH AMERICA

5.8.1.1 US

NGS sequencing technologies are being used predominantly in microbiome and metagenome sequencing. The increase in use is further supported by increasing advancements in genomics. Several federal agencies regulate NGS products, including the Food and Drug Administration (FDA), the Centers for Medicare and Medicaid Services (CMS), and the Federal Trade Commission (FTC).

CMS Regulations:

The CMS regulates clinical laboratories, including laboratories conducting clinical genetic testing, through its Clinical Laboratory Improvement Amendments of 1988 (CLIA) program. The CLIA established a certification that laboratories must pass to conduct clinical testing legally. The objective of the CLIA is to determine clinical testing quality, including verification of the procedures used and the qualifications of the technicians processing the tests.

FDA Regulations:

Until recent years, the FDA has practiced enforcement discretion for laboratory-developed tests (LDTs), which implies that LDTs are used in the clinic without the FDA's assessment of their analytical and clinical validity. However, due to rapid advances in NGS, the growing need for clinical genetic testing, the growth of direct-to-consumer (DTC) genomic testing, and concerns that unregulated tests pose a public health threat, the FDA is modifying its approach.

The FDA has developed new guidance to regulate NGS genetic tests and verify their analytical and clinical validity. The agency has also drafted guidance proposing a new regulatory framework for LDTs. However, FDA guidance only represents the FDA's current thinking on a topic and is not legally binding for the FDA or the parties it regulates. However, adhering to FDA guidance is beneficial because it can streamline the regulatory process. The draft guidance is listed below. Since they are in draft form, they are not currently being implemented.

List of FDA Draft Guidance on Genetic Tests:

- Framework for Regulatory Oversight of Laboratory Developed Tests (2014)
- FDA Notification and Medical Device Reporting for Laboratory Developed Tests (2014)
- Discussion Paper on Laboratory Developed Tests (2017)
- Next-generation Sequencing (NGS) Draft Guidance

- Use of Standards in FDA Regulatory Oversight of NGS-based In Vitro Diagnostics Used for Diagnosing Germline Diseases

In addition, the US FDA classifies medical devices into various categories. It reviews NGS instruments as Class II medical devices. The NGS manufacturer must submit a premarket notification or 510(k) to prove substantial equivalence to a legally marketed device. If there is no substantially equivalent device in the market, the manufacturer must secure premarket approval (PMA) from the FDA.

The three main classifications of medical devices by the US FDA are as follows:

TABLE 7 US FDA: MEDICAL DEVICE CLASSIFICATION

CLASS I DEVICES	CLASS II DEVICES	CLASS III DEVICES
▪ Low-risk medical devices, which are subject to the least regulatory control ▪ Exempt from premarket notification [510 (k)]	▪ Devices that pose a greater risk than Class I devices; require reasonable assurance of safety and effectiveness ▪ Require premarket notification [510 (k)], with exceptions for certain devices	▪ Devices that are used for sustaining human life and are important for preventing the impairment of human health ▪ These are the highest-risk devices and, hence, are subject to the highest level of regulatory control ▪ Require submission of clinical trial data to the US FDA and require Premarket Approval (PMA) from the FDA

TABLE 8 US: MEDICAL DEVICE REGULATORY APPROVAL PROCESS

DEVICE CLASSIFICATION	WAITING PERIOD AFTER SUBMISSION UNTIL APPROVAL IS GRANTED	VALIDITY PERIOD FOR REGISTRATION OF THE DEVICE	WHETHER REGISTRATION RENEWAL SHOULD BE STARTED IN ADVANCE	COMPLEXITY OF THE REGISTRATION PROCESS	OVERALL COST OF GAINING REGULATORY APPROVAL
Class I Device	1 Month	No Expiry Date	NA	Simple	Low
Class II Device	6–9 Months	No Expiry Date	NA	Moderate	Medium
Class III Device	18–30 Months	No Expiry Date	NA	Complex	High

5.8.1.2 Canada

Health Canada is a key regulatory authority for medical devices in the country. The Canadian Medical Devices Regulation (CMDR) is the key directive that governs the approval of medical devices (including NGS systems) in the country. In Canada, medical device manufacturers must first register and obtain a proper license from Health Canada to market their IVD devices. Health Canada has established a four-tier, risk-based classification system (Class I, II, III, and IV) for IVD products.

5.8.2 EUROPE

From May 2020, Europe has implemented new Medical Device Regulation (EU-MDR) guidelines to increase the safety of the medical devices used in the region. These guidelines require manufacturers to make significant data reporting and quality assurance changes. While new medical devices would be required to adhere to the new guidelines, existing medical devices would have to be recertified (in some cases, they would also have to be reclassified) by manufacturers.

In Europe, medical devices have been classified into four categories based on their function, risk to patients, and the manufacturer's intended use.

- Class I: Provide non-sterile or do not have a measuring function (low risk) and provide sterile and/or have a measuring function (low/medium risk); the MDR has added reusable surgical instruments to this group
- Class IIa (medium risk)
- Class IIb (medium/high risk)
- Class III (high risk)

However, with changes in MDR regulations, some of these medical devices have been reclassified. The new MDR regulations have 22 rules for classification, and the scope of medical devices has increased, including additional products.

Three Medical Device Directives mainly regulate the European market—the Medical Device Directive (93/42/EEC), the Active Implantable Medical Device Directive (90/385/EEC), and the In Vitro Diagnostic Medical Device Directive (98/79/EC). The European Medical Device Directive's Conformité Européenne (CE mark) approval is required for the distribution and marketing of any medical devices (including NGS systems) in the European (EU) region. This certification verifies that a device meets all regulatory requirements of the Medical Devices Directive (MDD). The manufacturer must be certified by the notified body to Annex II, V, or VI of the Medical Device Directives and comply with the requirements of those directives. A certified body conducts a technical review, and if all requirements are met, a CE clearance letter is issued, permitting the sale of a particular medical device in Europe. Also, manufacturers of medical devices must implement quality systems compliant with ISO 13485:2003.

5.8.3 ASIA PACIFIC

5.8.3.1 China

The National Medical Products Administration (NMPA), formerly the CFDA, is responsible for supervising and administering medical devices in the People's Republic of China (PRC). The NMPA classifies medical devices into Class I, Class II, and Class III based on the safety and effectiveness of the device.

TABLE 9 CHINA: CLASSIFICATION OF MEDICAL DEVICES

DEVICE CLASSIFICATION	POST-SUBMISSION APPROVAL (MONTHS)	LEVEL OF RISK	COMPLEXITY OF REGISTRATION	OVERALL COST OF GAINING REGULATORY APPROVAL	DEFINITION
Class I	<1	Low	Low	Moderate	Devices for which safety and effectiveness can be ensured through routine administration
Class II	16–24	High	High	Moderate-to-high	Devices for which further control is required to ensure safety and effectiveness
Class III	16–24	High	High	High	Devices that are implanted into the human body or that are used for supporting or sustaining life. They pose a significant risk to the human body and, hence, must be strictly controlled for their safety and effectiveness

NGS systems are classified as Class II devices as per NMPA rules. In China, IVD products (including NGS systems) and their components that are recognized internationally are classified as medical devices (MD), in vitro diagnostic reagents (including NGS reagents), and drugs. For an IVD system that includes an instrument and appropriate reagents, such as NGS, two product registration certificates must be acquired for market approval—medical device registration for instruments and IVD reagent registration for suited reagents.

5.8.3.2 Japan

The Ministry of Health, Labour and Welfare (MHLW) and the Pharmaceuticals and Medical Devices Agency (PMDA) govern the regulatory approval pathway to assess the safety and effectiveness of medical products in the country. Medical device manufacturers must abide by the PMDA. Obtaining PMDA approval for a new product is time-consuming (it can take approximately five years) and cumbersome. The compilation of foreign medical device manufacturers with PMDA is challenging due to the multilayered regulatory and language barriers. Hence, foreign manufacturers (with no local presence in Japan) need to appoint a Marketing Authorization Holder (MAH) to manage their device registration process and liaise with Japan's Pharmaceuticals and Medical Devices Agency.

5.8.3.3 India

According to the Central Drugs Standard Control Organisation (CDSCO), diagnostic kits are classified as notified diagnostic assays. Manufacturers must register these tests with the regulatory authorities and secure an import/manufacturing license. The remaining molecular diagnostic assays are termed non-notified diagnostic assays and need only an import license to be sold in India. The Drugs and Cosmetics Act of 1940 and the Drugs and Cosmetics Rules of 1945 are the major laws that regulate pharmaceuticals, medical devices, and diagnostics in India. The Ministry of Health & Family Welfare, through the Drug Controller General of India, is responsible for the improvement of public health by monitoring and regulating the quality, safety, and efficacy of drugs and devices through examination and investigation.

5.8.4 LATIN AMERICA

In some Latin American countries, such as Mexico and Argentina, medical device regulations are evolving, while in others, the regulations are static and, to some extent, opaque. Most Latin American countries consider premarket controls the primary feature for medical device regulation (including NGS). Medical device regulations in this region are driven by trade agreements, primarily Mercado Común del Sur (MERCOSUR), which seeks to create a common market like that in the EU. Medical goods and services can move freely among member countries. To register products in a Latin American country, a company must have an office in that country or have a local distributor (registered with the Ministry of Health) to register products on its behalf. For all Latin American registrations, a Free Sale Certificate (FSC) or a Certificate to Foreign Government (CFG) is required to confirm that a product is approved in the country of origin and can be exported without restriction.

5.8.5 MIDDLE EAST

Each member country of the Middle East has its regulatory system for medical devices in different maturity stages. Saudi Arabia has the most established regulatory framework for medical devices among Middle Eastern countries, followed by the UAE and Qatar. The regulatory systems of the UAE and Qatar are still in the early stages of development. The ministries of health in each of these countries are responsible for regulating medical devices. Most countries need overseas device manufacturers to appoint a local (or authorized) representative to facilitate the registration/regulation (in addition to distribution and sale) of their products in each country.

A marketing authorization from the Saudi Food and Drug Authority (SFDA) under Interim Regulation (1-8-1429) is required to sell medical devices in Saudi Arabia. Medical device manufacturers must obtain prior market authorization in reference markets such as Australia, Canada, the European Union, Japan, or the US. The device classification in respective reference markets will determine the classification of that device in Saudi Arabia. An authorized representative primarily manages the device registration in Saudi Arabia. The representative must be licensed with the SFDA.

5.8.6 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

TABLE 10 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS: GENOMICS, MICROBIOME RESEARCH, AND METAGENOMICS

REGION	COUNTRY	REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS
North America	US	- National Institute of Standards and Technology (NIST) - National Human Genome Research Institute (NHGRI) - Joint Genome Institute (JGI)
	Canada	- The Genomics Research and Development Initiative (GRDI) - Genome Canada
Europe	Germany	- DZIF German Centre for Infection Research - HELMHOLTZ-ZENTRUM FÜR INFEKTIONSFORSCHUNG GMBH
	UK	- UK Research and Innovation (UKRI) - Microbiology Society
	France	- Paris Center for Microbiome Medicine (PaCeMM) FHU - Microbiome Foundation France
	Italy	- Microbiome Research Hub Italy - Council for Agricultural Research and Economics (CREA)
Asia	Spain	- The Centre for Genomic Regulation (CRG) - National Centre for Genomic Analysis (CNAG-CRG) - The Higher Council for Scientific Research - The Institute for Research in Biomedicine (IRB)
	Japan	- Human MetaGenome Consortium Japan (HMGJ) - Japan Agency for Medical Research and Development
	China	- Agricultural Genomics Institute at Shenzhen (AGIS) - National Genomics Data Centre China
	India	- Department of Biotechnology (GENOME India and Microbiome) - International Life Sciences Institute India (ILSI)
	South Korea	- National Research Foundation of Korea (NRF) - Korea Biotechnology Industry Organization (KoreaBIO)
	Australia	- National Research Infrastructure for Australia (Australian Microbiome Initiative) - Microbiome Research Centre (MRC)
Latin America	Brazil	- Brazilian Society of Genetics (Sociedade Brasileira de Genética in Portuguese) - The São Paulo Research Foundation (Fundação de Amparo à Pesquisa do Estado de São Paulo in Portuguese)

Middle East & Africa	Saudi Arabia	▪ The Saudi Society of Medical Genetics (SSMG) ▪ King Faisal Specialist Hospital and Research Centre's (KFSH&RC) Genetics department ▪ Medical Genetics Programme ▪ Center of Innovation in Personalized Medicine (CIPM)
	South Africa	▪ South African Medical Research Council (SAMRC) ▪ South African Society for Human Genetics ▪ Cancer Association of South Africa (CANSa) ▪ Hyrax Biosciences

Source: Government Websites and MarketsandMarkets Analysis

5.9 PRICING ANALYSIS

The pricing of microbiome sequencing services can vary depending on several factors, including the specific sequencing technology used, the depth of sequencing coverage, the number of samples, the data analysis and interpretation required, and the infrastructure capacity of the service provider.

TABLE 11 AVERAGE SELLING PRICE, BY SERVICE AND TYPE

SERVICE PROVIDER	REGION/COUNTRY	TYPE	SERVICE	SAMPLE SIZE/ CUSTOMER TYPE	PRICE RANGE
Dalhousie University Integrated Microbiome Resource (IMR)	North America	Amplicon Sequencing (16S/18S/ITS Short Amplicons)	Library Preparation + Sequencing	50 to 200 Samples	USD 20–30
			Sample Preparation (DNA Extraction)	50 to 200 Samples	USD 15–20
			Standard Bioinformatics Pipeline	50 to 200 Samples	USD 500–1,000
		Metagenome Sequencing	Library Prep. + Sequencing	Academic Customers	USD 100–150
			DNA Extraction	Academic Customers	USD 15–20
			Standard Bioinformatics Pipeline	Academic Customers	USD 1,000–1,200
Genotypic Technology	Asia Pacific	Metagenomic Analysis	Library Preparation + Sequencing	Per Sample	USD 20–25